Physical Chemistry CUME, Apr. 8, 2016

Chemical Kinetics

Q1. Each of the reactions shown below consists of two "elementary" steps characterized by rate constants k_f and k_r. Determine the relative concentration of reactant(s) and product in terms of k_f and k_r as $t \to \infty$ (i.e., equilibrium condition) in each case. What is the order of reaction associated with each elementary step?

(a) $\mathrm{A} \underset{k_r}{\overset{k_f}{\rightleftharpoons}} B$ (10 points)

(b) $\mathrm{A} + \mathrm{B} \underset{k_r}{\overset{k_f}{\rightleftharpoons}} C$ (15 points)

Q2. The complex reaction shown below involves three elementary steps with rate constants $\mathrm{k_f}$, $\mathrm{k_r}$, and $\mathrm{k_{cat}}$ for the forward (binding), reverse (dissociation), and catalytic reactions, respectively. Assuming $\mathrm{k_f}, \mathrm{k_r} \gg \mathrm{k_{cat}}$ (i.e. binding and dissociation occur much faster than the catalysis), an instantaneous equilibrium can be assumed between the intermediate species ES (enzyme-substrate complex) and the reactants (free enzyme E and substrate S) at any given time:

$$\mathrm{E} + \mathrm{S} \underset{k_r}{\overset{k_f}{\rightleftharpoons}} ES \xrightarrow{k_{cat}} E + P$$

(a) Defining $K_d \equiv k_r/k_f$, find the concentration of the intermediate species ES in terms of K_d, [E] (free enzyme concentration) and [S] (substrate concentration)? (10 points)

(b) Since the total enzyme concentration does not change, the initial enzyme concentration $[E]_0$ is equal to the sum of the concentrations of the free and bound enzymes ([E] and [ES], respectively) at any time. Find an expression for [ES] in terms of $[E]_0$, [S], and K_d. (30 points)

(c) How is the product formation rate (i.e., $\frac{d}{dt}[P]$) related to the concentration of intermediate species ES? (10 points)

(d) Show that: $\frac{d}{dt}[P] = \frac{k_{cat}[E]_0\,[S]}{K_d+[S]}$ (5 points)

(e) What is the maximum rate of product formation (V_{max}) as the enzyme becomes saturated by substrate? Find the substrate concentration [S] at which the rate of product formation is $\frac{1}{2}V_{max}$. (20 points)